

# HENRY KOEHN

Cedar Falls, IA • 319-883-1322 • henrykoehn1@gmail.com

## EDUCATION

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**The University of Iowa — B.S.E., Computer Science and Engineering**

*Iowa City, IA*

**Anticipated May 2027**

## TECHNICAL SKILLS

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**Languages:** Python, C#, C++, SQL

**Machine Learning & Data Science:** Pandas, NumPy, Scikit-Learn, Jupyter Notebook

**AI Tooling:** Copilot Studio / AI Agents, Power Automate

**Cloud & Infrastructure:** Azure, Docker, Linux

**DevOps & Automation:** Jenkins, Ansible, Chef, Git, GitHub Actions

**Monitoring:** Dynatrace

**Frameworks:** .NET, Blazor, Entity Framework

## PROFESSIONAL EXPERIENCE

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**Site Reliability Engineering Intern**

*Mastercard*

**June 2026 – August 2026**

*O'Fallon, MO*

- Built and integrated a Copilot AI agent with a Power Automate flow to automatically generate Post-Incident Reports (PIRs) from Microsoft Teams call transcripts of emergency TRT (Technical Recovery Team) incident calls, eliminating a manual documentation process; solution approved for production rollout at Mastercard.
- Supported enterprise API Gateway infrastructure as part of a Site Reliability Engineering (SRE) team responsible for reliability and operational health.
- Built Dynatrace dashboards to monitor real-time health and status metrics across gateway server environments, improving infrastructure visibility for engineering teams.
- Utilized Jenkins, Ansible, and Chef to support build automation and configuration management workflows.
- Participated in daily operational handoffs between U.S. and India engineering teams to maintain service continuity across time zones.

**Research Assistant — Sinapse Lab**

*University of Iowa*

**February 2026 – Present**

*Iowa City, IA*

- Containerized a deep-learning tractography pipeline (CoRNN) using Docker, enabling reproducible execution across Linux research environments.
- Resolved dependency management, GPU compatibility, and multi-tool integration challenges involving FSL, ANTs, and MRtrix3.
- Validated containerized workflows across multiple Linux distributions and documented deployment and troubleshooting procedures for research use.

**Software Engineering Intern**

*Casey's*

**May 2025 – August 2025**

*Ankeny, IA*

- Developed and maintained production accounting applications using C#, .NET, SQL, Docker, Blazor, and Azure in an Agile development environment.
- Contributed to the redesign and modernization of a sales reconciliation platform for a Fortune 500 organization.
- Queried and managed relational databases using Entity Framework and SQL across multiple application environments.

## PUBLICATIONS

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- Co-Author, "Machine Learning-Based Prediction of Infant Hypoglycemia" — manuscript in preparation with University of Iowa Stead Family Children's Hospital clinicians; planned submission to Hospital Pediatrics, September 2026.

## PROJECTS

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**Machine Learning Pilot Research**

- Collaborated with University of Iowa Stead Family Children's Hospital clinicians to develop ML models predicting infant hypoglycemia.
- Built Python-based data preprocessing and model development workflows utilizing Pandas, NumPy, and Jupyter Notebooks.
- Contributed technical implementation for a pilot study evaluating predictive approaches for early identification of at-risk patients.

**Pairs Trading Back Tester Application**

- Built a back testing tool using C++ and Python to test stock trading strategies on historical data.